

ARYAN GUPTA

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ACADEMIC QUALIFICATIONS

- **Masters of Science in Statistics**, Hindu College, University of Delhi | CGPA: 8.5 / 10 **May 2026**
- **Bachelors of Science (Hons) in Statistics**, Kirori Mal College, University of Delhi | CGPA: 8.8 / 10 **May 2023**
- **Class XII, CBSE Board**, D.L. D.A.V. Model School, Pitampura, Delhi | Percentage: 94 % **March 2020**
- **Class X, CBSE Board**, D.L. D.A.V. Model School, Pitampura, Delhi | Percentage: 90.6 % **March 2018**

INTERNSHIPS

Summer Intern, Institute of Economic Growth | Delhi, India **June 2025 – August 2025**

- Analyzed **600K+** NFHS-5 records in Python (pandas, scipy) to map hemoglobin deficiency and anemia prevalence across 5+ demographic segments; applied **CLT & Z-scores**, rigorous **data cleaning and outlier detection** for statistical robustness.
- Surfaced **region-wise** and **pan-India** public health patterns from complex survey data, translating findings into evidence-based policy recommendations for government stakeholders.

Data Analyst Intern, Health & Family Welfare Vertical – NITI Aayog | Delhi, India **July 2023 – September 2023**

- Analyzed performance data from **500+** district hospitals across FY21–22 to identify operational gaps; findings directly informed national health policy recommendations.
- Built **Excel dashboards** (pivot tables, charts) tracking **15+** KPIs for ministerial briefings, supporting strategic planning in collaboration with senior policymakers.

ACADEMIC PROJECTS

Cookie Cats A/B Testing Analysis | Python, Statistics, Bayesian Inference, Machine Learning **May 2026 – June 2026**

- Analyzed gate position impact on player retention ($n = 90,189$) using **SRM check**, **Mann-Whitney U**, **2-proportion Z-test**, and **Kaplan-Meier survival analysis**; Gate 30 significantly outperformed Gate 40 (Bayesian $P(\text{win}) = 99.9\%$).
- Segmentation revealed effect concentrated in moderate-to-engaged cohort (11–200 rounds, $p = 0.0001$); **XGBoost** retention model (**ROC-AUC = 0.889**) confirmed engagement volume as dominant D7 predictor over gate version.

Airline Customer Churn Prediction | Python, Machine Learning, Classification **November 2025 – December 2025**

- Predicted customer churn on **100K+** records (**25 features**) using **Logistic Regression**, **Random Forest**, and **XGBoost**; best model achieved **F1-score = 0.96** with online boarding experience and inflight Wi-Fi service as top churn drivers.
- Performed **EDA** addressing **multicollinearity** and **missing values**; applied **hyperparameter tuning** and evaluated models across ROC-AUC, F1-score, and Confusion Matrix to select final classifier.

Spotify Song Popularity Prediction | Python, Machine Learning, Regression **October 2025 – November 2025**

- Predicted track popularity on **114K** Spotify records (**21 features**) using **Linear**, **Ridge**, **Lasso**, and **Ensemble Models**; best model achieved $R^2 = 0.70$ with danceability and valence identified as dominant popularity drivers.
- Applied **correlation analysis**, **hypothesis testing**, and categorical encoding to engineer features; derived ranking-based insights for playlist curation and music marketing optimization.

TECHNICAL SKILLS

Languages & Tools: Python, R, SQL, Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn, Advanced Excel, MS Word, Power Point

Statistics & Machine Learning: A/B Testing, Bayesian Inference, Survival Analysis, Statistical Analysis, Hypothesis Testing, Predictive Modelling, Classification, Regression, Ensemble Methods, Feature Engineering, Hyperparameter Tuning, Data Wrangling

Analytics: EDA, Statistical Inference, Data Visualization, Business Insights & Data Storytelling

LEADERSHIP & ACHIEVEMENTS

Student Coordinator, Udaan – the socio-cultural cell | Department of Statistics, UoD **September 2024 – May 2026**

- Led community initiatives (cloth collection drive with NGOs, blood donation camp with **200+ registrations**); coordinated large scale departmental events, demonstrating **initiative, teamwork, planning, and time management**.

President, Department of Statistics | Kirori Mal College, UoD **September 2022 – May 2023**

- Led 20+ member teams to organize seminars and annual fest with **100+ participants**; planned and executed department trip for 30+ students, showcasing **leadership, communication, and problem-solving**.

Academic Merits & Credentials:

- **AIR 135** in IIT JAM MS 2024, **AIR 21** in ISI MS QMS 2024, **100%** in Mathematics (Class XII).

Certifications:

- Data Analytics Simulation (Deloitte), Data Analysis with Python (IBM Skills Build).